

1 ResNet as a Discrete Feynman Path Integral

We interpret ResNet as a discrete-time dynamical system inspired by the Feynman path integral formulation. Instead of viewing the model as a simple stack of layers, we treat it as a superposition of exponentially many computational paths.

The final output is expressed as:

$$\text{Final Output} = \sum_{\text{all paths}} e^{-S_{\text{path}}} u(x_0) \quad (1)$$

where:

- S_{path} represents the action (energy cost),
- Each path corresponds to a combination of residual and skip transitions.

2 Layer-Level Architecture

Each residual block is defined as:

$$u(x, t+1) = u(x, t) + F(u(x, t), W_t) \quad (2)$$

where:

- $u(x, t)$ is the feature map at depth t ,
- F is the residual function,
- W_t are learnable parameters.

Residual Block Structure

For image-based research, each block consists of:

$$\begin{aligned} \text{Input} &\rightarrow \text{Conv}(3 \times 3) \rightarrow \text{BN} \rightarrow \text{ReLU} \\ &\rightarrow \text{Conv}(3 \times 3) \rightarrow \text{BN} \\ &\rightarrow \text{Add (Skip Connection)} \rightarrow \text{ReLU} \end{aligned}$$

Mathematically:

$$F(u_t) = W_{2,t} \sigma(W_{1,t} u_t) \quad (3)$$

Thus, the full update becomes:

$$u_{t+1} = u_t + W_{2,t} \sigma(W_{1,t} u_t) \quad (4)$$

This is equivalent to the Euler discretization of:

$$\frac{\partial u}{\partial t} = F(u, t) \quad (5)$$

3 Path Integral Interpretation

Due to skip connections, each layer has two possible transitions:

- Skip path $\rightarrow$ multiply by 1
- Residual path $\rightarrow$ multiply by W_t

For N layers:

$$\text{Total Paths} = 2^N \quad (6)$$

Each path is defined by:

$$p = (s_1, s_2, \dots, s_N) \quad (7)$$

where

$$s_t = \begin{cases} 1 & \text{if skip} \\ W_t & \text{if residual} \end{cases} \quad (8)$$

The contribution of a path is:

$$u_{\text{path}} = u_0 \prod_{t=1}^N s_t \quad (9)$$

Thus, the final output becomes:

$$u_N = \sum_{p \in \mathcal{P}} u_0 \prod_{t=1}^N s_t \quad (10)$$

4 Hamiltonian-Based Energy Formulation

To incorporate the physical interpretation, define the Hamiltonian:

$$H(p) = \frac{1}{2}(\sigma^2 p^2 - ibp - c) \quad (11)$$

where:

- p = path momentum,

- σ^2 = diffusion coefficient,
- b = drift term,
- c = potential term.

The action of a path:

$$S_{\text{path}} = \int_0^T H(p(t)) dt \quad (12)$$

Discretized form:

$$S_{\text{path}} = \sum_{t=1}^N H(p_t) \quad (13)$$

The final output:

$$u_N = \sum_{\text{paths}} e^{-S_{\text{path}}} u_0 \quad (14)$$

5 Complete Network Architecture for Research

Input Layer

- Image size: $224 \times 224 \times 3$
- Initial Convolution: 7×7 , stride 2
- Max Pooling

Residual Stages (ResNet-34 Style)

Table 1: Residual Stage Configuration

Stage	Blocks	Channels
Stage 1	3	64
Stage 2	4	128
Stage 3	6	256
Stage 4	3	512

6 Modified Path-Integral ResNet

6.1 Energy Regularization

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \sum_{\text{paths}} S_{\text{path}} \quad (15)$$

This suppresses high-energy paths and promotes low-energy paths.

6.2 Learnable Energy Scaling

$$u_{t+1} = u_t + \alpha_t F(u_t) \quad (16)$$

where

$$\alpha_t = e^{-H_t} \quad (17)$$

6.3 Diffusion-Controlled Depth

Let depth be normalized time:

$$t = \frac{k}{N} \quad (18)$$

As $N \rightarrow \infty$, the model approaches:

$$\frac{\partial u}{\partial t} = \sigma^2 \Delta u + b \nabla u - cu \quad (19)$$

7 Training Procedure

Step 1: Forward Pass

Compute residual transitions normally.

Step 2: Energy Computation

$$H_t = \frac{1}{2}(\sigma^2 \|W_t\|^2 - bW_t - c) \quad (20)$$

Step 3: Action

$$S = \sum H_t \quad (21)$$

Step 4: Loss

$$\text{Loss} = \text{CrossEntropy} + \lambda S \quad (22)$$

8 Why This Architecture is Powerful

1. Gradient Stability

Skip path represents a zero-energy trajectory, ensuring stable gradient flow.

2. Implicit Ensemble Effect

ResNet behaves as an exponential ensemble of shallow subnetworks.

3. Physics-Guided Regularization

Energy controls:

- Overfitting
- Weight explosion
- Noise propagation

9 Research Extension Directions

- Stochastic Path Sampling
- Quantum-Inspired Attention
- Energy-Based Pruning
- Diffusion-ResNet Hybrid

10 Final Conceptual Summary

$$\text{ResNet} = \text{Discrete Feynman Path Integral over Residual Transformations} \quad (23)$$

Where:

- Skip = identity path
- Residual = perturbation path
- Depth = time
- Weights = local Hamiltonian
- Final output = weighted sum of computational trajectories